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Asia Thematic Strategy

Why AI remains foremost

Equity Macro Research

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Price action in Asia in 2026 can be characterized at the highest level as AI vs non-AI – with lesser differentiation between other sectors/markets. We expect this pattern to carry through into 2027, with the **outcome for AI stocks determining not only allocation alpha but also overall market beta**. Indeed, AI stocks in Asia are now >50% of index weight (*Figure 1*) and are likely to contribute ~70% of earnings growth in 2027 (*Figure 2*; similar to 2026). AI exposure in Asia is primarily in the hardware layer, both in the global/US ecosystem (Taiwan, Korea and some China) and the China ecosystem, with the vast majority of the earnings dependency and index weight in the former. The **economics of this hardware layer are currently very positive** – with supply-demand tightness (expected at least through 2028) leaving Asian firms with significant pricing power to go with volume growth – driving ecosystem net income to over $1tn by 2028. But having transitioned from a volume to a pricing/margin trade late last year, the key question for these stocks now is about the duration of the spending cycle. And as the JPM Tech equity research team has highlighted, the duration depends largely on how well AI is monetized above the hardware layer – at the hyperscalers and, more importantly, at their customers – AI models and applications. **At the hyperscaler layer, ROI at this point is very strong (>30%,** *Figure 3*), and we believe the funding constraints can be overcome (FCF should start to improve by 2028, *Figure 4*) – which should keep the current momentum in AI capex intact. Power is another constraint – but also being addressed. Monetization at the model/application layer is the main focus of markets at present and incoming information on this front will continue to drive sentiment about the broader AI ecosystem. We believe there are **sufficient pathways for the model layer to deliver on monetization** (approaching $300bn in ARR by end-2026; *Figure 7*; and anything better than 2x annual growth from there should be good enough). For now, the developments remain positive – agents and cybersecurity (*Figure 6*) are presently driving token growth (*Figure 5*), while regulation, safety and geopolitics are the risks to consider. Further out, even higher level questions need to be answered about value capture, competitive advantages, economic displacement, social structures, financial stability, etc., but the evidence we have at present is not sufficient to conclude on these in any definitive way. Overall, we believe it remains important to be **open-minded about the extent of AI upside** and reassess conditions from time to time, instead of subscribing to definitive timelines and end-states. As such, **we remain OW Tech/Taiwan/Korea in our regional allocation recommendations and AI remains the biggest among our preferred themes**.

- **Economics of the hardware layer:** The main driver here is **ongoing demand exceeding existing and upcoming supply –** leaving the Asian hardware ecosystem with substantial pricing power amid continued strong growth in volumes. This should bring ecosystem net income to over $1tn by 2028 and FCF to ~$700bn (*Figure 22-23*). (1) But investors continue to worry about the impact of an eventual end of this supply-demand imbalance, leading them to **persistently underprice the longevity** of the cycle. Taking the memory sector as an example – JPM analysts show that current market valuations are pricing in very bearish end-of-cycle conditions. (2) There will always be segments

See page 23 for analyst certification and important disclosures, including non-US analyst disclosures.

within the ecosystem where **supply-demand tightness is worsening or easing**. This remains a key differentiating factor within the ecosystem and a focus area for L/S investors. JPM Tech Research team provides regular updates across the various segments. The most persistent shortages are likely to remain in advanced logic semis and packaging, semi equipment, ABF substrates, high-end CCL materials, HBM and high-speed optics. (3) The earnings at Korean and Taiwanese Tech companies are shaping up to be **material even at a macro level** (household incomes, government revenues, banking liquidity, etc.) with a variety of spillover effects to consider (*Figure 8*).

- **Economics of the hosting/hyperscaler layer:** This mainly relates to the cost of depreciation and opex vs the revenue generation from renting out compute. Currently, this **economics is extremely favorable**. If datacenters are depreciated over five years on aggregate, at current rental rates, this equates to about **30% ROI** on conservative assumptions. (1) As a result, the hyperscaler spending cycle has shown **little sensitivity to interest rates** because, even with higher yields, the incremental financing burden is modest relative to the scale of earnings. Emergence of compute futures will also help broaden financing availability as collateral valuation becomes more bankable. (2) The temporal dislocation between **cash outflows** (at the time of build) and inflows (only after datacenter is operational and rented out) means hyperscalers are running out of self-funding (cash flows) and need external financing. For now, there is enough cash in the hardware layer to support this, and hyperscaler access to credit markets remains stable. And if datacenter ROI is so high, and capex growth is slowing (from >100% in 2026 to <50% in 2027 as per JPMe), then FCF should not be perpetually negative. Indeed, our simple math suggests that the **AI-only FCF hole may be closing already by 2028**. (3) We believe the real hyperscaler constraints will remain in component and labor procurement, politics/permitting and access to power – which will dictate how fast and where datacenters come up.
- **Economics of the model/application layer:** This is the top of the funnel for the AI ecosystem – which needs to generate enough revenues to pay itself and all the layers below eventually. But this is also where the economics are hardest to predict, given the lack of information, changing business models, regulatory setup and competitive landscape, as well as evolving use cases of AI. (1) We used to be concerned about the monetization here in late 2025 as the pricing model at that time (per seat/user charge) did not appear to scale well, and our belief (then and now) is that **AI businesses are not platform businesses** of the internet era and do not benefit from the same network effects – and as such lack significant moats. This means that pursuing profit-less growth for scale and market share is not a good strategy. **Since late 1Q 2026, the pricing model in AI has shifted to "per token" and token subsidies have faded** – which not only scales very well but has unlocked the upside in AI revenues and shortened the path to profitability. (2) At a high level, **revenue now is a product of token volume and token price**. Most of the time, token pricing is deflating but token volumes are growing very fast. Beyond this, for the sustainability of the model layer economics itself, we need to also consider the cost of producing a token, and the magnitude of resources needed for continued model training and R&D. (3) Over this summer, as enterprise token budgets overran, we have seen a rapid optimization of token usage and a substantial **shift in the mix of token demand towards cheaper models –** even as overall token demand growth continues apace. As such, aggregate token pricing has deflated rapidly since mid-year (pressuring model-maker economics), but this **may end up being a one-off adjustment**. (4) **Token volume is perhaps the most important variable**. Volumes are partially impacted by token pricing, but primarily by expanding use cases (new capabilities) and user base (adoption). Model capabilities show no signs of slowing yet, and Ramp AI data suggest enterprise adoption of AI is still rising. Many investors worry that after "reasoning" in 2025 and "coding" in 2026, there isn't an obvious block-buster use case for 2027. But there are many candidates – **agentic** (which, despite the early enthusiasm around OpenClaw, has only recently started to go mainstream) and **cybersecurity** appear to be

the most imminent ones, while usage in robotics and metaverse also appears to be sizable in the future. (5) Finally, resources needed for continued model training and R&D will depend on the **competitive landscape** (regulatory treatment of open-source models, ease of cloning capabilities, etc.) as well as how much AI models can contribute to improving themselves (**recursive self-improvement**). (6) Overall, we believe that there are sufficient pathways for the model layer to deliver on monetization. One **plausible modest scenario for 2027** involves aggregate token pricing still deflating by 70% (down to less than a third), overall token consumption growing 10x again, leaving **AI ARR to 3x to almost $1tn**. If inference gross margins are ~50% on aggregate and inference as a share of compute remains ~60% (like in 2026), this translates into a AI model layer operating profits run rate (after inference costs and training spend) of ~$167bn, with a hyperscaler ARR at end 2027 of ~$833bn (which is not very different from the numbers implied by current GPU rental rates and estimated capex). **Visibility on this sort of economics should be very positive for stock prices across the ecosystem**.

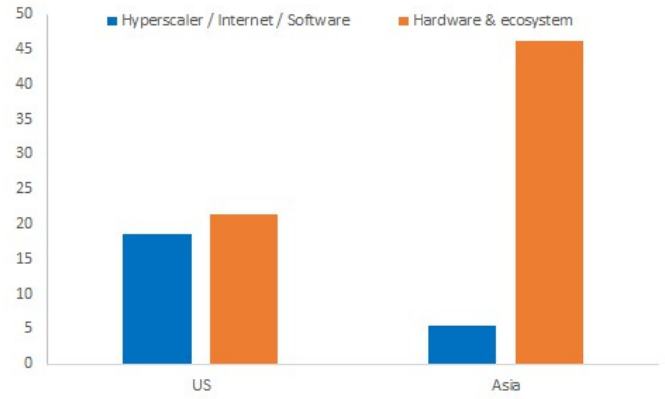
- **Beyond monetization:** So far, AI's interaction with the economy has largely been in the form of AI capex – which for the most part adds activity and jobs to the economy. But AI outputs may soon become large enough to have **meaningful macroeconomic consequences**. A key question is how the mix pans out between AI primarily generating efficiency gains that replace existing activity and jobs, vs creating new products, services, and sources of demand. Another important distinction from the internet era is that AI companies may not benefit from the same network-effect moats, which could lead to a more balanced distribution of value across hardware, hosting, models, enterprises, and consumers. Much of the **surplus may ultimately accrue to end users** rather than to any single corporate layer. However, that outcome becomes more complicated if the **same end users are also facing labor-market displacement**, potentially intensifying wealth concentration and the K-shaped economy. Finally, **regulation, AI-safety concerns, cybersecurity risks, and U.S.-China competition** could all become increasingly important drivers of deployment timing, model-release cadence, and eventually semiconductor demand.
- **Key pushbacks and risks:** (1) A material **slowdown in token consumption** growth remains the most important risk. Monetization risks (in the absence of clear moats) have not gone away, but strong token demand can keep that at bay. (2) The other risk to monetization is **commoditization –** across AI models and AI hardware, as increasing amounts of supply come online. (3) More than funding and regulation, we see **power generation** as an important constraint to datacenter buildout. But power equipment production is ramping up, and more AI DC sites are moving to behind-the-meter (BTM) power. (4) Regulation and safety-driven shifts in the **competitive landscape** could go either way. (5) **Shifts in relative supply-demand balance** and pricing power (organically or due to step-changes in Tech) can be risks for specific segments of the stack. (6) Given the concentration risks and increasingly interlinked financing within the AI ecosystem, a **failure of any of the large companies** could threaten to cascade across the ecosystem – with implications for broader financial stability and the global economy. (7) **Price action in 1999-2000** shows internet stocks (top of the funnel) peaked well before the downstream picks-and-shovels. While this parallel was more ominous in July-Aug, it remains a risk. (8) **Valuations and positioning** are challenging to interpret. Traditional valuation metrics fail in the very-unique current conditions. Positioning is similarly tricky, and it appears to us that in many cases LO investors aren't able to be as OW as they would like. And HF positioning in AxJ Tech is substantially lighter now vs the highs of June.
- **Top exposures in Asia AI:** Taiwan Tech (TSMC, ASE, MediaTek, Wiwynn, ASPEED, Chroma, Unimicron), Korea Tech (Samsung, Hynix, SEMCO), Japan Tech (Tokyo Electron, Advantest, Nittobo), HK/China Tech (ASMPT, NAURA, Iluvatar, Zhipu) and China Internet (Alibaba).

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Figure 1: The weight of AI-related stocks (mostly in hardware) has increased to >50% of AxJ

% of MXASJ



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 2: Almost 70% of AxJ earnings growth in 2026 and 2027 driven by AI

Net income in US$ bn

USD bn	2023	2024	2025	2026 E	2027 E	2028 E
Information Technology	96	135	194	624	884	1,038
Industrials	99	113	117	155	179	206
Consumer Discretionary	76	109	104	111	140	171
Financials	467	518	556	600	639	680
Communication Services	37	51	62	75	83	93
Health Care	18	23	21	30	34	41
Materials	48	45	55	108	124	141
Consumer Staples	40	49	47	46	57	62
Utilities	33	44	49	46	49	53
Energy	82	85	69	98	94	100
Total	**996**	**1,170**	**1,275**	**1,893**	**2,284**	**2,586**
Increase		174	104	619	391	301
US AI	**59**	**108**	**158**	**560**	**791**	**918**
Increase		49	50	403	231	127
Share		28%	48%	65%	59%	42%
China AI	**49**	**53**	**66**	**96**	**127**	**159**
Increase		4	13	30	30	33
Share		2%	13%	5%	8%	11%

Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 3: Hypothetical sensitivity of AI ROI for a hyperscaler

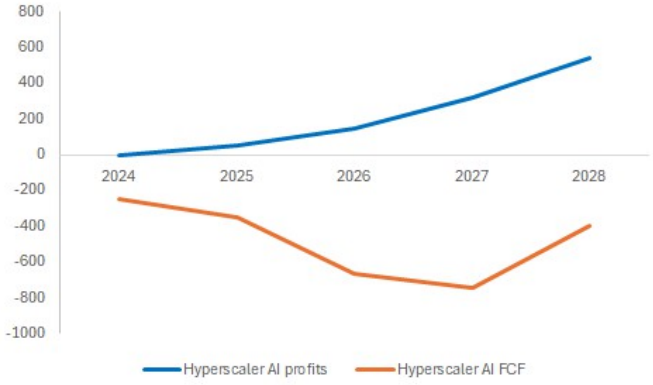
Indicative numbers for a B200 based datacenter. Assumes 550k GPUs per GW, $1.5bn annual operating expense per GW and $60bn in upfront capex (year 0) that starts to pay from year 1, and a 90% utilization rate

		Depreciation horizon				
		2yr	3yr	4yr	5yr	10yr
Rental Rate ($/GPU/hr)	3	-46%	-23%	-10%	-1%	14%
	4	-34%	-11%	2%	10%	23%
	5	-23%	0%	13%	20%	31%
	6	-12%	11%	23%	30%	39%
	7	-3%	21%	32%	39%	47%
	8	7%	30%	42%	47%	55%

Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 4: Hyperscaler AI FCF hole should start to close by 2028

Capex spend based on JPM Cloud Capex Database. Assumes 1GW costs over 2024-28 as $35bn, $45bn, $60bn, $65bn and $70bn. Assumes revenues per GW from 2024-28 vintage hardware at $15bn, $20bn, $25bn, $25bn and $25bn with a 5yr depreciation/useful-life schedule. Assumes opex at 15% of depreciation costs



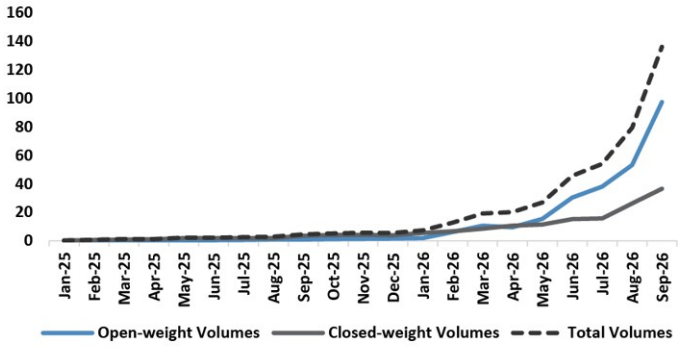
Source: Bloomberg Finance L.P., J.P. Morgan.

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Figure 5: Growth in token demand is one of the most important determinants of economics at the model layer

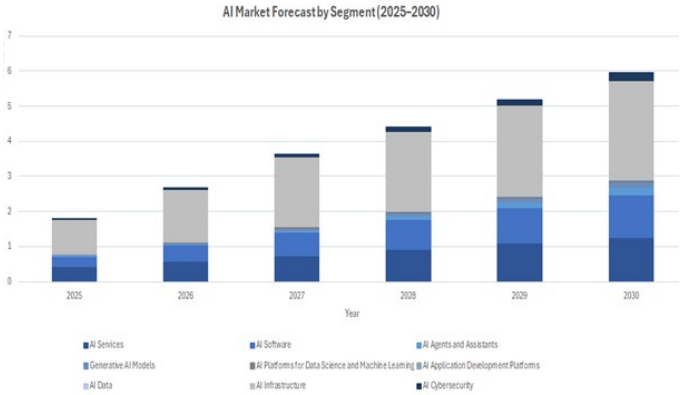
Weekly tokens in trillion. See here



Source: Openrouter, J.P. Morgan estimates.

Figure 6: Cybersecurity will likely be a big growth driver in 2027; JPM estimates an incremental $430bn+ 3yr TAM

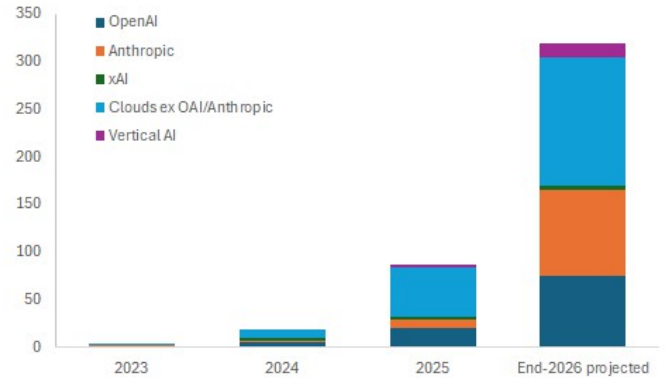
Market size estimate in US$ tn. See more here



Source: Gartner report, J.P. Morgan.

Figure 7: Total AI revenue run rate (ARR) at year-end could reach over $300bn, >3x from end-2025

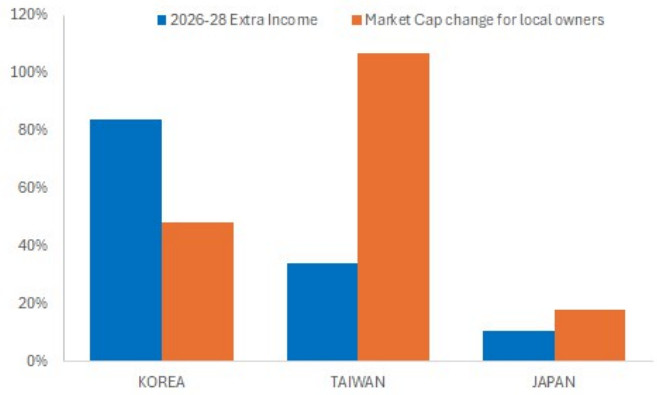
$bn



Source: Bloomberg Finance L.P., news reports (see here and here), Exponential View, J.P. Morgan.

Figure 8: Hardware company profits/stock valuation gains are becoming macro-relevant now – especially in Korea and Taiwan

% of GDP



Source: Bloomberg Finance L.P., J.P. Morgan.

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AI still the dominant theme in markets

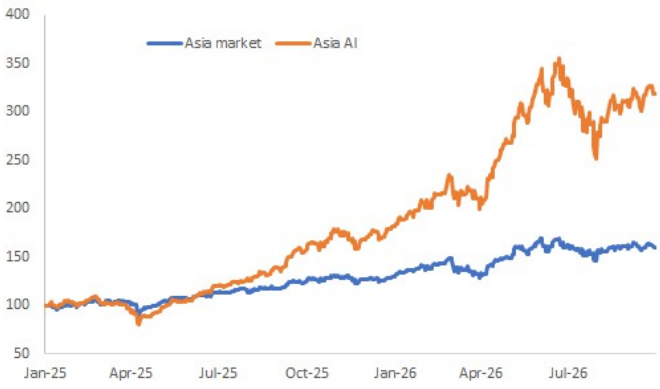
Price action in Asia in 2026 can be characterized most appropriately as AI vs non-AI. **AI has been the dominant theme driving both allocation alpha and market beta**, with limited differentiation between other sectors/markets and index upside largely driven by gains in AI stocks. We **expect this pattern to carry through into 2027**. Macro conditions for other sectors are likely to remain challenging in an environment of elevated rates and further widening of an already K-shaped economy.

We see the AI theme being driven by a combination of (1) fundamental strength, (2) incoming narrative updates driving long-term expectations about the fundamentals, and (3) positioning. While positioning in such a dominant theme naturally gets crowded from time to time, leading to pullbacks like in July, fundamental strength in the Asian AI ecosystem remains unrelenting and should be able to overcome such periods of volatility. As such, **AI stocks in Asia are likely to again contribute ~70% of the earnings growth in 2027** (similar to 2026).

With this growth, **AI-exposed stocks in Asia now comprise >50% of index weight in MSCI Asia ex-Japan**. Similarly, Korea and Taiwan weight in MSCI EM overall has now reached close to 50%.

Figure 9: AI stocks have vastly outperformed the market despite the July drawdown

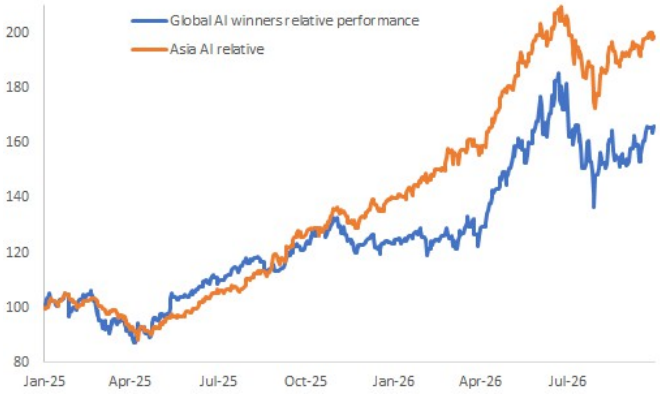
Absolute performance



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 10: Relative performance of AI stocks is picking up again – in Asia and in the US

Relative performance



Source: Bloomberg Finance L.P., J.P. Morgan.

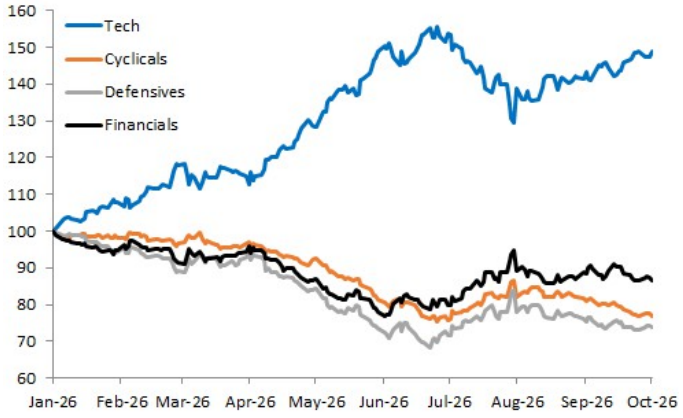
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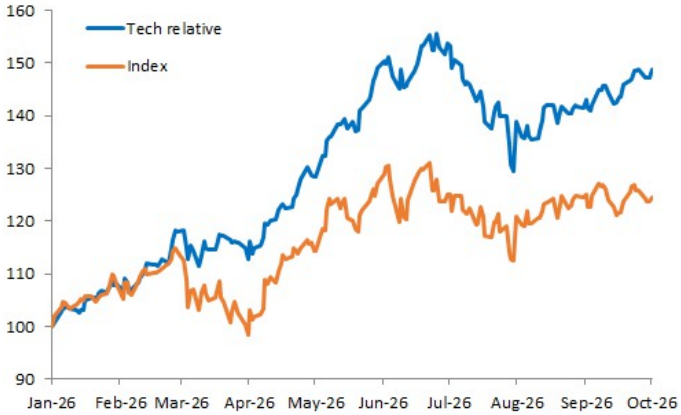
Figure 11: In Asia, Tech has been the main outperforming sector – with little differentiation between others

Relative performance



Source: Bloomberg Finance L.P., J.P. Morgan.

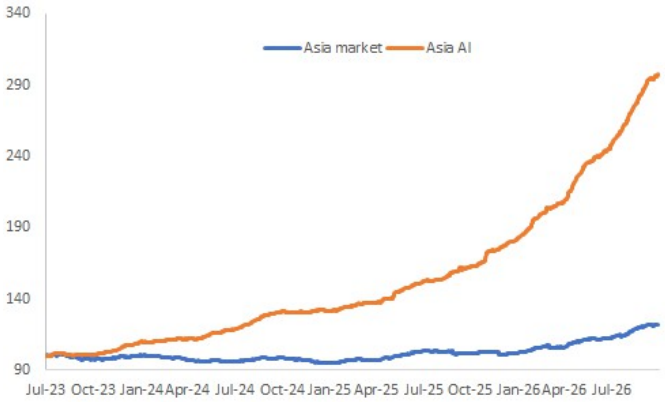
Figure 12: Asia index upside has also largely coincided with Tech outperformance



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 13: Sales estimates for AI stocks are vastly outpacing the market

Estimated 12m fwd sales



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 14: Almost 70% of AxJ earnings growth in 2026 and 2027 driven by AI

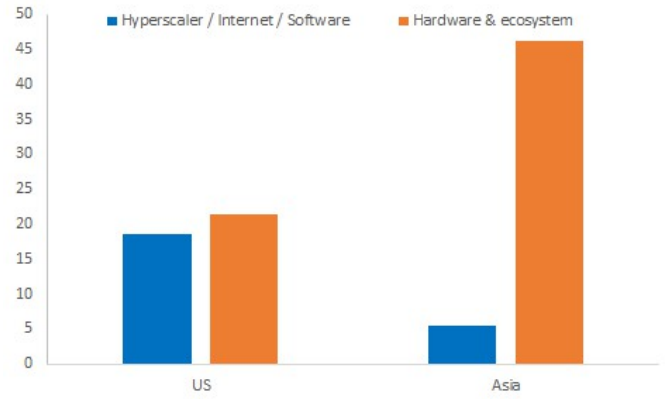
Net income in US$ bn

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Increase		4	13	30	30	33
Share		2%	13%	5%	8%	11%

Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 15: As such, the weight of AI related stocks (mostly in hardware) has increased to >50% of AxJ

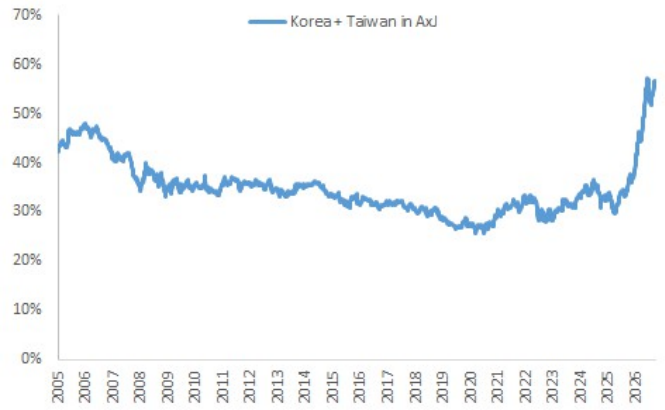
% of MXASJ



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 16: Similarly, the weight of Korea + Taiwan now exceeds 50% of AxJ

% of MXASJ



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 17: J.P. Morgan preferred stocks in the Asia AI ecosystem

Name	Ticker	Industry Group	Market cap ($mn)	3m ADV ($mn)	Price (lc)	YTD Perf	Fwd P/E	P/B	Fwd ROE	JPM Rating
TSMC	2330 TT	Semiconductors & Semicondu	2,102,339	2,101	2,585	66.8	19.4	10.42	40.5	OW
Samsung	005930 KS	Technology Hardware & Equip	1,171,691	7,273	268,500	123.0	4.1	2.74	50.1	OW
Hynix	000660 KS	Semiconductors & Semicondu	939,486	9,391	1,723,000	164.3	3.9	4.67	72.3	OW
Alibaba	9988 HK	Consumer Discretionary Distri	267,052	1,304	105	-26.2	13.8	1.47	9.6	OW
Mediatek	2454 TT	Semiconductors & Semicondu	243,365	1,154	4,835	238.1	38.4	18.19	41.0	OW
Advantest	6857 JP	Semiconductors & Semicondu	190,822	2,046	41,290	110.3	38.3	28.67	61.0	OW
Tokyo Electron	8035 JP	Semiconductors & Semicondu	184,684	1,329	12,500	82.1	29.8	13.26	38.7	OW
ASE	3711 TT	Semiconductors & Semicondu	102,630	381	732	192.2	27.9	8.27	26.6	OW
SEMCO	009150 KS	Technology Hardware & Equip	89,429	1,312	1,604,000	527.8	42.1	11.41	23.2	OW
Unimicron	3037 TT	Technology Hardware & Equip	67,173	718	1,305	493.2	36.5	16.75	37.4	OW
NAURA	002371 CH	Semiconductors & Semicondu	66,653	1,060	616	34.2	43.2	11.02	20.0	OW
Zhipu	2513 HK	Software & Services	43,241	939	696	-	-	6.00	34.3	OW
Wiwynn	6669 TT	Technology Hardware & Equip	39,279	317	2,240	49.0	13.5	27.19	48.6	OW
Chroma	2360 TT	Technology Hardware & Equip	32,007	184	2,400	209.7	39.6	30.82	57.8	OW
Aspeed	5274 TT	Semiconductors & Semicondu	23,148	158	17,750	168.9	58.4	104.20	87.8	OW
Iluvatar	9903 HK	Semiconductors & Semicondu	9,974	138	99	-	93.4	10.83	8.0	OW
ASMPT	522 HK	Semiconductors & Semicondu	9,325	63	175	126.3	30.3	4.21	12.4	OW
Nittobo	3110 JP	Capital Goods	4,376	113	3,675	80.1	27.0	3.77	13.2	OW

Source: Bloomberg Finance L.P., J.P. Morgan. Note: Prices as of 7 October 2026.

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Economics of the hardware layer

As the chart above shows, unlike that in the US, the AI ecosystem in Asia largely comprises hardware companies (in Taiwan, Korea, Japan and China) producing semiconductors and associated products/equipments for the global AI datacenter buildout. The economics of these firms largely relate to **ongoing demand exceeding existing and upcoming supply** – leaving the Asian hardware ecosystem with **substantial pricing power** amid continued strong growth in volumes.

Once the period of supply shortage passes us (in some future steady state), hardware economics will likely revert closer to what it has been in the cloud era (relatively thin margins and steady growth rates), but the timing of such a moderation remains unclear. Indeed, having transitioned from a volume to a pricing/margin trade late last year, the **key question for these stocks now is about the duration of the spending cycle**. And as the JPM Tech equity research team has highlighted (see here), the duration depends largely on how well AI is monetized above the hardware layer – at the hyperscalers and, more importantly, at their customers – AI models and applications. In such uncertain conditions, **markets have (naturally) persistently underpriced this longevity**. Taking the memory sector as an example – JPM analysts show that current market valuations are pricing in very bearish end-of-cycle conditions for memory stocks.

In the meantime, hardware companies' earnings (and the increases in stock price-related national wealth) are so large that they have become **relevant at a macro level**. In Korea, aggregate market earnings over 2026-28, as per consensus estimates, will be 80% of GDP in excess of the 2025 baseline. And the market-cap increase for local investors adds another ~50% of GDP.

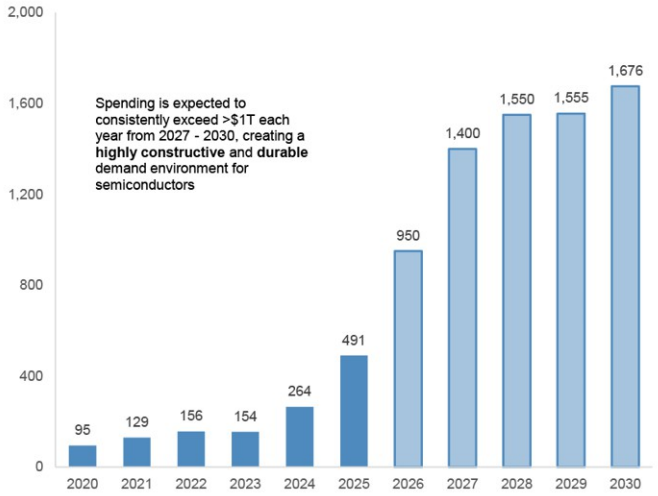
Finally, intra-sector differentiation remains driven by relative supply-demand tightness and pricing power. Gross margin estimates across a broad sample of AI hardware companies have increased ~8% on average since Sep-25. The broad expectation for **2027–2028 is continued AI-driven tightness**, but not uniformly across the Asian hardware supply chain. The most persistent shortages are likely to remain in **HBM, advanced logic and packaging, semiconductor manufacturing equipment, ABF substrates, high-end CCL materials, and high-speed optics**. Commodity NAND (DRAM is still needed in AI servers) is the clearest area where supply-demand conditions may normalize earlier (but even this may be delayed if memory makers continue shifting capacity towards HBM), while power and cooling could remain constrained, mainly at the project and component level rather than as a universal global shortage.

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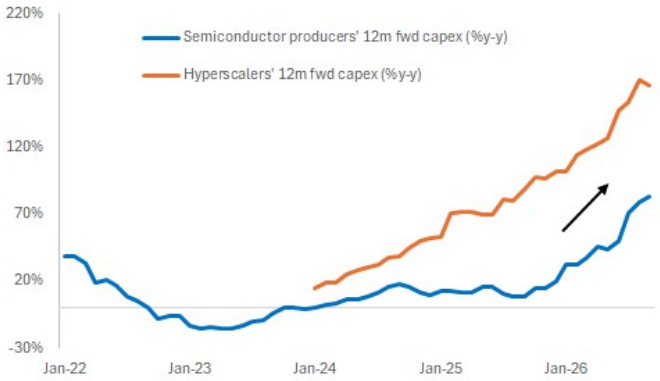
Figure 18: JPM estimates annual cloud capex to rise to $1.4tn in 2027 – continuing to drive demand for hardware

US$ bn. As at end-August 2026. See here



Source: Bloomberg Finance L.P., J.P. Morgan estimates.

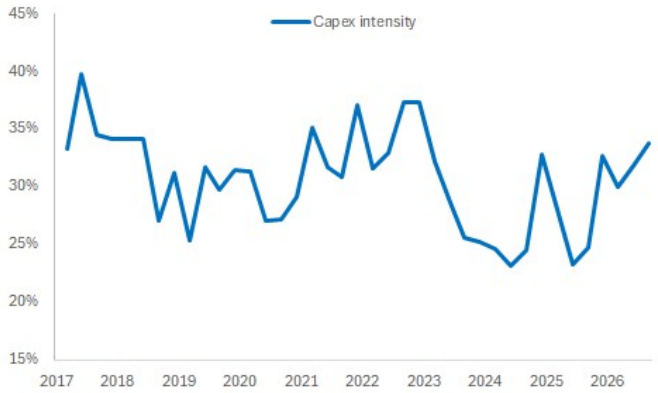
Figure 19: Compared to the growth rate of hardware demand (AI capex), the growth rate of supply capex is slower (and growth of installed supply capacity is even slower)



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 20: In fact, capex intensity is not really high so far

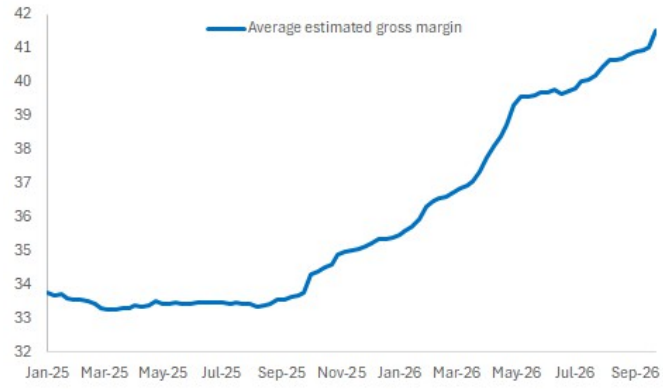
Annualized quarterly capex / net fixed assets for the 13 largest semiconductor producers



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 21: With this supply-demand mismatch, margins for Asian AI hardware names continue to rise

Average of 12m forward gross margin across 25 AI semi / tech hardware companies in Asia



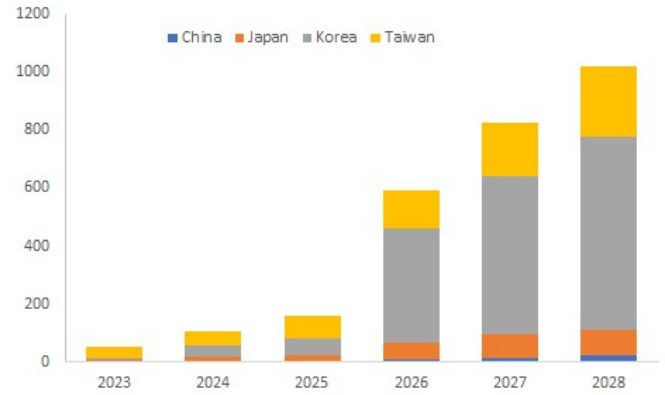
Source: Bloomberg Finance L.P., J.P. Morgan.

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Figure 22: Overall, net income of AI ecosystem companies in Asia is expected to exceed $1tn by 2028

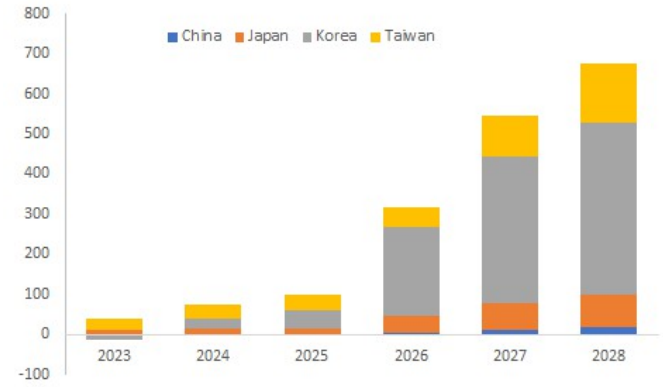
Actuals and consensus estimates. US$ bn



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 23: Similarly, free cash flow for AI ecosystem companies is expected to reach $700bn by 2028

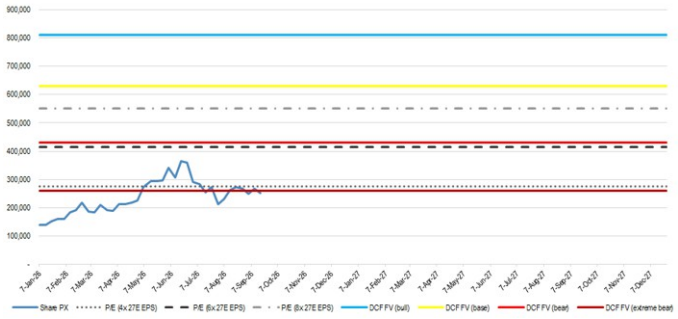
Actuals and consensus estimates. US$ bn



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 24: In contrast, market pricing of capex durability is not very aggressive – for example, in Samsung

See valuation and DCF details here



Source: Bloomberg Finance L.P., J.P. Morgan estimates.

Figure 25: In contrast, market pricing of capex durability is not very aggressive – for example, in Hynix

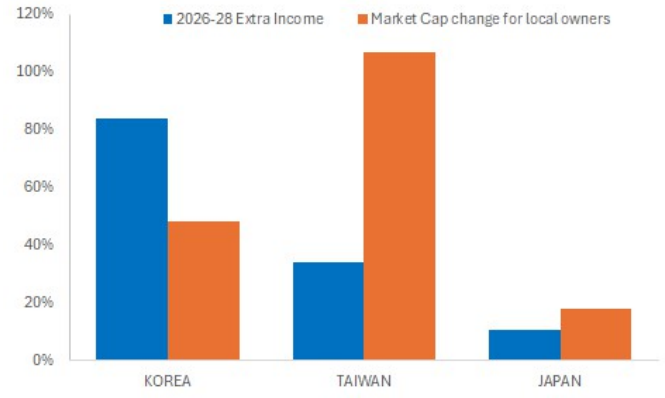
See valuation and DCF details here



Source: Bloomberg Finance L.P., J.P. Morgan estimates.

Figure 26: Hardware company profits/stock valuation gains are becoming macro-relevant now – especially in Korea and Taiwan

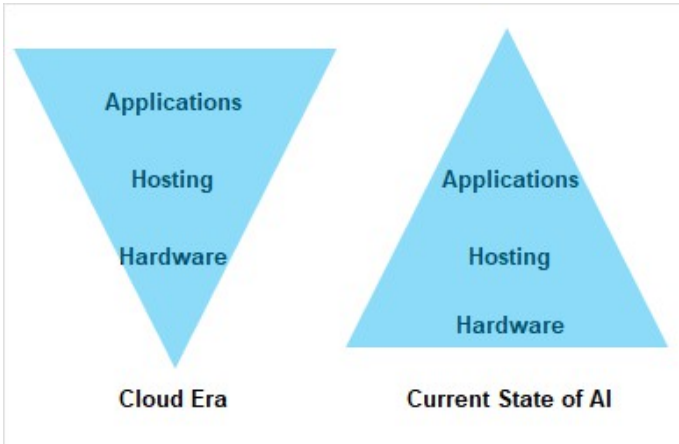
% of GDP



Source: Bloomberg Finance L.P., J.P. Morgan estimates.

Figure 27: Once the supply shortage runs its course, the steady state value capture is unlikely to look like the current one

Indicative distribution of value capture across the stack



Source: Bloomberg Finance L.P., J.P. Morgan estimates.

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Economics of the hosting layer

For the hyperscalers (hosting layer), the business model is building and then renting out compute (to internal or external users). The economics relate to the initial capex, duration of useful life, operating expenses, and the rental revenue generation. Currently, this **economics are extremely favorable**. If datacenters are depreciated over five years on aggregate, at current rental rates, **building a B200 datacenter equates to about 30% ROI**, leaving plenty of room to overcome financing costs and inflation in components.

As a result, the hyperscaler spending cycle has shown little sensitivity to interest rates because, even with higher yields, the incremental financing burden is modest relative to the scale of earnings and many can still fund a meaningful portion of buildouts internally while maintaining strong balance sheets. As a result, **capex decisions are being driven less by the marginal cost of debt and more by strategic urgency**: AI workloads are growing quickly, the opportunity cost of delaying capacity is high, and "time-to-power/time-to-serve" has become a competitive variable. With data center lead times, supply-chain constraints (chips, racks, cooling), and power and interconnect availability all acting as gating factors, hyperscalers are prioritizing securing sites and capacity now rather than waiting for a lower-rate environment that may arrive after the critical window for AI leadership has already moved on.

An additional nuance to consider is the temporal dislocation between cash outflows (at the time of build) and inflows (only after datacenter is operational and rented out). For this reason, hyperscalers are running out of self-funding (cash flows) and need external financing, at least for a while. But if datacenter ROI is so high, and capex growth is slowing (from >100% in 2026 to <50% in 2027 per JPMe), then FCF should not be perpetually negative for hyperscalers. Indeed, our **simple math suggests that the FCF hole may be closing already by 2028**. This assumes (1) hyperscaler capex of $1tn, 1.4tn and 1.55tn in 2026-28 as per the JPM Cloud Capex Database. (2) 1GW costs over 2024-28 of $35bn, $45bn, $60bn, $65bn and $70bn. (3) Revenues per GW from 2024-28 vintage hardware of $15bn, $20bn, $25bn, $25bn and $25bn with a 5yr depreciation/useful-life schedule. (4) Opex at 15% of depreciation costs. With these assumptions, we also note that in 2027, AI capex growth is ~50%, AI GW (real) growth is ~25%, and installed GW capacity growth is ~60%.

Similar to the debate about steady-state hardware value capture, a longer-term question is whether hyperscalers will eventually make utility margins, cloud margins or software margins? Software margins look untenable (this requires a moat that is absent at the hosting layer), but cloud margins (~30%) are likely feasible.

Unlike interest rates and funding availability, we believe the **real hyperscaler constraints will remain in procurement, politics and power**. The procurement feeds directly to the hardware economics (and shortage) discussed above. The politics is about datacenter moratoriums and political movements to slow down AI development (although this is not quite a hot-button issue yet). As for power, the move is clearly towards behind-the-meter power generation to supply datacenters. Over time, datacenter construction outside the US (and even in space) may be alternatives.

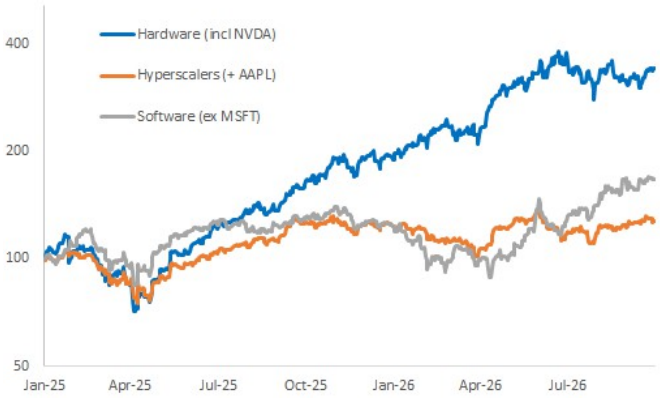
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Figure 28: In the US, hardware performance is recovering, software has gained from the early-2026 crash and hyperscalers are also finding some traction

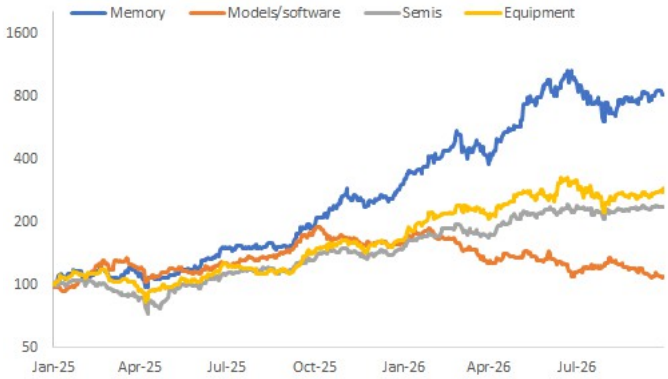
Absolute performance, log scale



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 29: Performance of various major segments of the Asia AI ecosystem

Absolute performance, log scale



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 30: Spot rental rates for B200 GPUs have been rising in recent months

US$ / hour spot rates offered by neoclouds



Source: Bloomberg Finance L.P., Silicon Data, J.P. Morgan.

Figure 31: Spot rental rates for even older (H100) GPUs are not only holding up, but actually rising

USD / hour spot rates offered by neoclouds

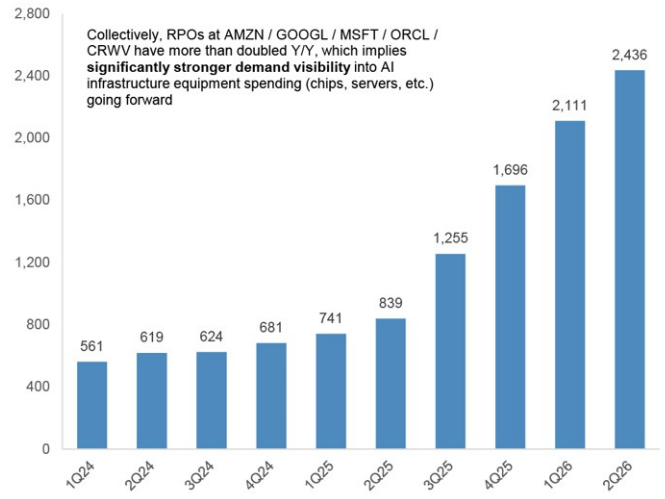


Source: Bloomberg Finance L.P., J.P. Morgan.

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Figure 32: And the hyperscalers continue to have a very large backlog of compute orders

Remaining performance obligations (RPO) from compute agreements



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 33: Hypothetical ROI sensitivity for a hyperscaler

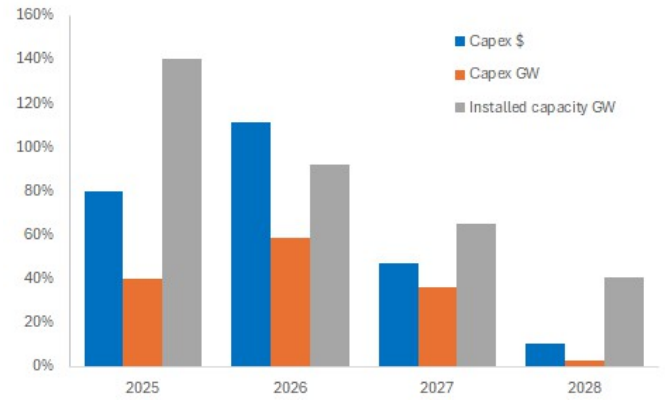
Indicative numbers for a B200 based datacenter. Assumes 550k GPUs per GW, $1.5bn annual operating expense per GW and $60bn in upfront capex (year 0) that starts to pay from year 1, and 90% utilization rate

Rental Rate ($/GPU/hr)	Depreciation horizon 2yr	3yr	4yr	5yr	10yr
3	-46%	-23%	-10%	-1%	14%
4	-34%	-11%	2%	10%	23%
5	-23%	0%	13%	20%	31%
6	-12%	11%	23%	30%	39%
7	-3%	21%	32%	39%	47%
8	7%	30%	42%	47%	55%

Source: Bloomberg Finance L.P., J.P. Morgan estimates.

Figure 34: Installed capacity of AI datacenters is likely only growing ~60% annually from 2027

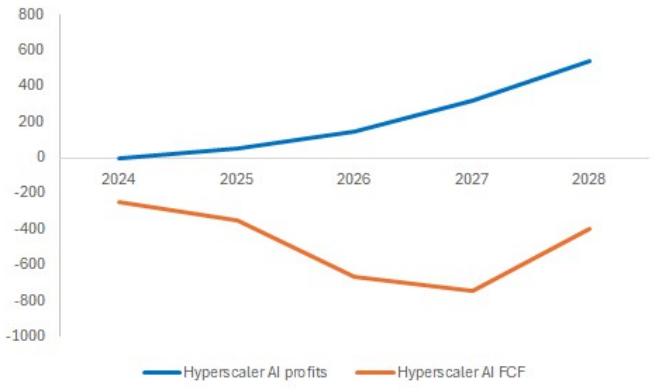
Capex spend based on JPM Cloud Capex Database. Assumes 1GW costs over 2024-28 as $35bn, $45bn, $60bn, $65bn and $70bn.



Source: Bloomberg Finance L.P., J.P. Morgan estimates.

Figure 35: Hyperscaler AI FCF hole should start to close by 2028

Capex spend based on JPM Cloud Capex Database. Assumes 1GW costs over 2024-28 of $35bn, $45bn, $60bn, $65bn and $70bn. Assumes revenues per GW from 2024-28 vintage hardware of $15bn, $20bn, $25bn, $25bn and $25bn with a 5yr depreciation/useful-life schedule. Assumes opex at 15% of depreciation costs.



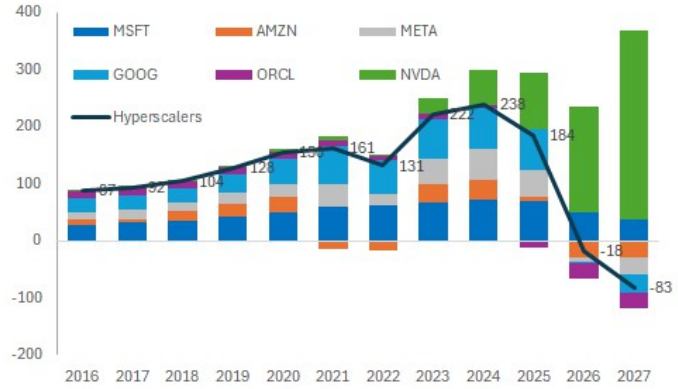
Source: Bloomberg Finance L.P., J.P. Morgan estimates.

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Figure 36: More than 100% of the FCF decline at hyperscalers ends up at NVDA

$ bn, free cash flows and estimates. calendar-year adjusted.



Source: Bloomberg Finance L.P., J.P. Morgan estimates.

Figure 37: IG hyperscalers' credit spread has widened but remains very modest

bps



Source: Bloomberg Finance L.P., J.P. Morgan estimates.

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Economics of the model layer

This is the top of the funnel for the AI ecosystem – which needs to generate enough revenues to pay itself and all the layers below eventually. We used to be concerned about the economics here in late 2025 as the pricing model at that time (per seat/user charge) did not appear to scale well, and our belief (then and now) that AI businesses are not platform businesses of the internet era and do not benefit from the same network effects – and as such lack significant moats. This means that pursuing profit-less growth for scale and market share are not good strategies. Since late 1Q 2026, the pricing model in AI has shifted to "per token" – which scales very well and has unlocked the upside in AI revenues and shortened the path to profitability.

At a high level, the revenue now is a product of two variables – **token volume** and **token price**. In traditional microeconomic terms, the token supply curve is relatively flat but rapidly rising, and the demand curve is reasonably steep. Thus, most of the time, token pricing is falling and token volumes are growing very fast. Beyond this, for the sustainability of the model layer economics itself, we need to consider two other variables – the **cost of producing a token**, and the magnitude of **resources needed for continued model training** and R&D.

Unfortunately, this layer is where the economics are hardest to model, given lack of information and predictability across these four variables. We need to do a lot of guesswork/leap-of-faith on use-cases, and deal with changing business models, competitive landscape, technology shifts and evolving regulatory setup.

Token cost is perhaps the most predictable variable here, and a function of hardware and model efficiency as well as power cost. And we can say with reasonable confidence that the quality-adjusted cost of generating tokens will continue to deflate. For instance, nVidia's GB300 system is over 30x more performant than the HGX200, and the new generation of VR systems continue that trend. Similarly, models continue to improve in inference efficiency and software optimizations remain possible.

Token price is a somewhat-visible variable, even if less predictable. In a free market for tokens, the quality-adjusted supply-vs-demand determines token prices. Effectively, **token pricing is an amalgamation of various different markets balancing supply-demand of varying qualities of tokens**. This ranges from frontier, and typically closed-source and pricier models to competent but cost-effective, typically open-source models. This quality-grading became very stark over the summer as major enterprise users started to hit their full-year token budget targets. What followed was a rapid optimization of token usage to match the nature of the task to the quality (and hence price) of the token – and thus a substantial shift in the mix of token demand – even as overall token demand growth continued apace. **As such, aggregate token pricing has deflated rapidly since mid-year (pressuring model-maker economics), but this may end up being a one-off adjustment.** As long as the aggregate value (usefulness) of a token remains higher than its aggregate price (which we currently believe is comfortably, and increasingly, the case), demand is likely to be strong, keeping token price from deflating too fast. In fact, even open-source models are exploring revenue-sharing agreements with hyperscalers to secure funding for future training/R&D.

Token volume is perhaps the most important variable. Volumes are partially impacted by token pricing, but primarily by expanding use cases (new capabilities) and user base

(adoption). While many investors worry that after "reasoning" in 2025 and "coding" in 2026, there isn't an obvious block-buster use case for 2027. But there are many candidates – agentic (which, despite the early enthusiasm around OpenClaw, has only recently started to go mainstream) and cybersecurity appear to be most imminent, while usage in robotics and metaverse also appear to be sizable in the future. Note that the **vast majority of token generation growth is coming through such hardware and software improvements rather than increases in installed datacenter capacity**. Tokens are growing at 10-20x annually, while datacenter capacity (in GW) is growing in the 50-70% range at most. This also means that new datacenters coming online next year will need to be ~15x performant (hardware+software) to deliver on 10x token growth.

Resources needed to stay competitive depend on the perceived upside from training/ R&D spending vs downside from model performance stagnation (rapid commoditization). On the former, while model companies continue to invest in training and R&D and reap benefits from it (in terms of improved model capabilities or cost competitiveness), it is also the case that training spend has been growing slower than overall AI capex and as such training share of compute has been falling in recent years (to ~40% now, according to AMD). The main change underway here is the potential emergence of **recursive self-improvement** (RSI) where AI models themselves do AI research to improve future versions, potentially making model improvement cheaper. On the latter, the key variable is the **regulatory stance** – if at some point open-source or models developed in adversarial nations are deemed unsafe from a security standpoint, the competitive landscape could change.

On aggregate, we believe that while there is **still some need for funding at the model layer** (preferably equity funding given the lack of collateral at this layer), there are **sufficient pathways for the model layer to deliver on monetization.** Indeed, we note that on aggregate AI ARR (annualized revenue runrate) is on course to triple to over $300bn. And while news flow in this context remains hectic (model launches/ capabilities, incidents, self regulation and political posturing), the broad brush of incoming information remains positive for now. JPM's Tech equity research team notes: "*Our back-of-envelope calculation indicates that a hypothetical model vendor could make $20bn-40bn in annual revenue per GW selling AI tokens, much higher than the $10bn per GW per year that existed in 2025 (based on OpenAI CFO Sarah Friar's comments). We believe that the surge of vertical AI companies that have reportedly reached $100M+ ARR in a very short span of time (80+companies already based on a Sapphire Venture report) is also a testament to the value creation downstream of selling Compute.*"

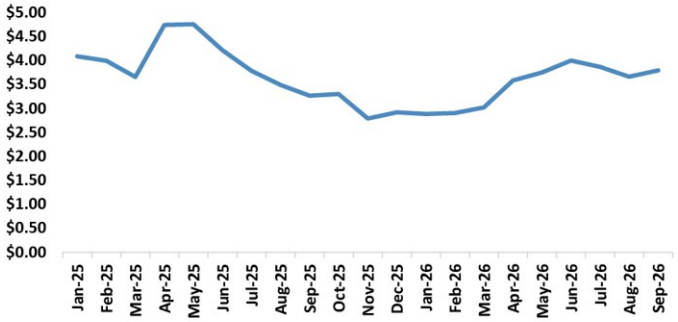
A plausible modest scenario for 2027 involves aggregate token pricing still deflating by 70% (down to less than a third), overall token consumption growing 10x, leaving AI ARR to again 3x to almost $1tn. If inference gross margins are ~50% on aggregate and inference as share of compute remains 60% (like in 2026), this translates into an AI model layer operating profits run rate (after inference costs and training spend) of ~ $167bn, hyperscaler ARR at end 2027 of ~$833bn (which is not very different from the numbers implied by current GPU rental rates and estimated capex). Visibility on this sort of economics should be very positive for stock prices across the ecosystem. Indeed, if AI ARR simply doubles every year, it will likely exceed annual capex by 2029.

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Figure 38: Average token pricing doesn't tell the full picture

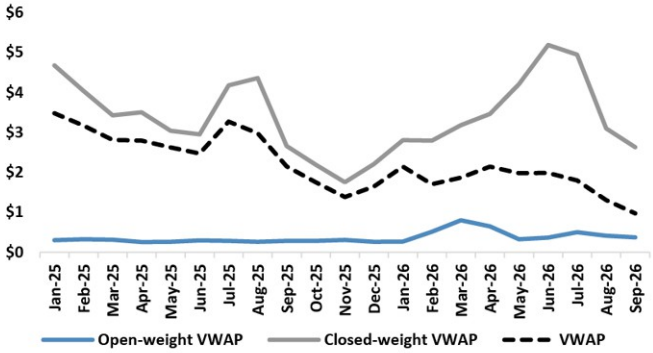
$ per million tokens. See here



Source: OpenRouter, J.P. Morgan estimates. *Includes models without observable volumes on OpenRouter.

Figure 39: Weighing by model usage, weighted average token price has deflated from $3 to $1 over the past year (-66%)

$ per million tokens. See here



Source: OpenRouter, J.P. Morgan estimates.

Figure 40: Silicon Data's token pricing index currently shows a similar price point

$ per million tokens.



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 41: But the cost of producing tokens is deflating too – Newer generations of hardware are orders of magnitude efficient in token generation

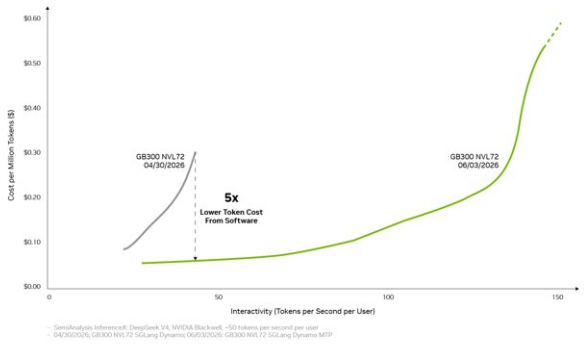
See here

Term	NVIDIA Hopper (HGX 200)	NVIDIA Blackwell (GB300 NVL72)
Intelligence Delivered	71.7% (Real-world coding benchmark)	
Cost Per Token	$4.2 per 1M tokens	$0.12 per 1M tokens
Cost Per Run 20B* rollout tokens x Cost per Token	~$84K	~$2.4K
Frequency of Runs	Multiple post-training runs in agentic era. Lowest cost per token makes frequent runs affordable.	
Intelligence Per Dollar (per $1,000)	0.85 (baseline)	29 (higher)

Source: nVidia blog.

Figure 42: Similarly, software / model updates are also lowering token production costs

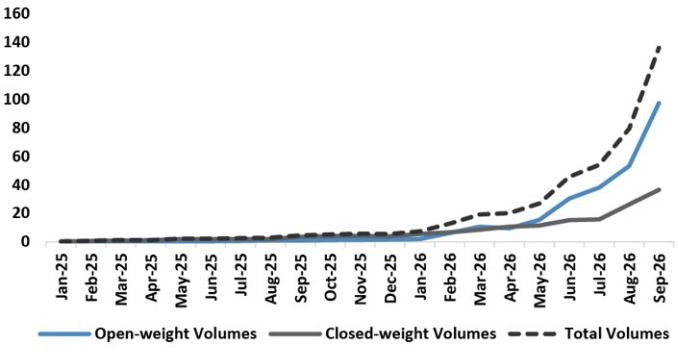
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Source: nVidia blog.

Figure 43: Growth in total token consumption then determines the $ gross profit

Weekly tokens in trillion. See here



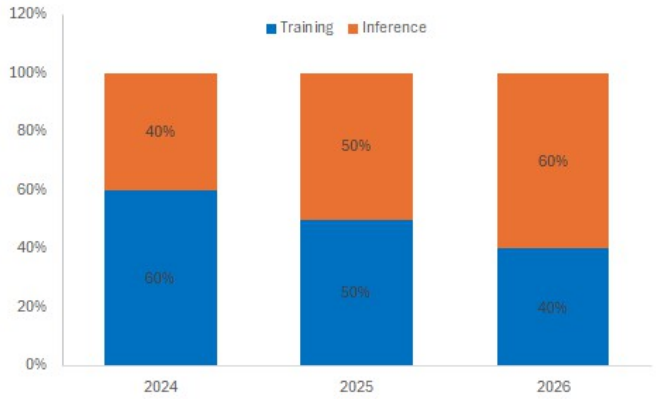
Source: OpenRouter, J.P. Morgan estimates.

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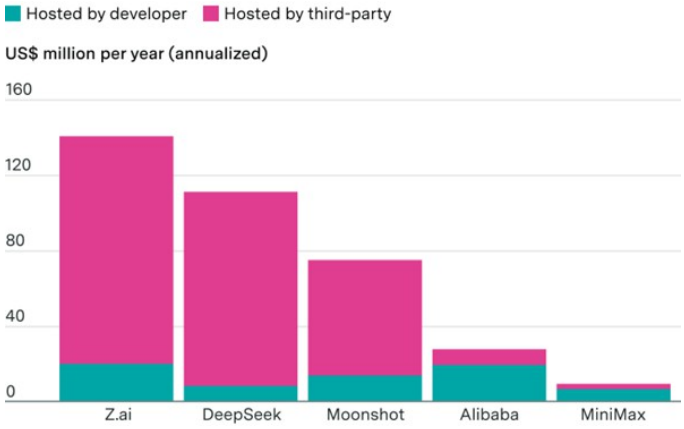
Figure 44: Inference is now about 60% of total token workloads



Source: AMD.

Figure 45: Most Openrouter spending on Chinese models goes to third-party hosts, not to model developers

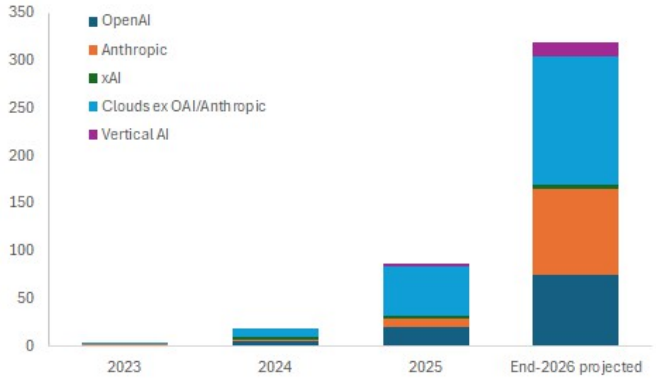
Estimated spend, annualized from eight sample days between late Aug to mid Sep 2026



See here
Source: Openrouter, Epoch AI.

Figure 46: Total AI revenue run rate (ARR) at year-end could reach over $300bn, >3x from end-2025

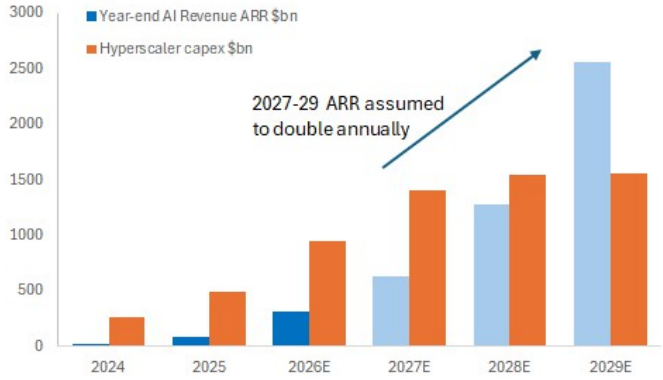
$bn



Source: Bloomberg Finance L.P., news reports - here and here, Exponential View, J.P. Morgan.

Figure 47: Even if ARR was only doubling from here, revenues would exceed capex by 2029

$bn. Capex spend based on JPM Cloud Capex Database.



Source: Bloomberg Finance L.P., J.P. Morgan estimates.

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Longer-term issues

While the market debate on AI is likely to remain about the monetization issues discussed above for now, given the salience and far-reaching consequences of the development of AI, it is important to start considering questions that may be in play further out.

(1) The first of these is about **economic displacement**. So far, AI's meaningful interaction with the economy has largely been in the form of AI capex – which for the most part adds activity and jobs to the economy. On the other hand, AI revenues this year ($300bn ARR by year-end, 0.25% of global GDP) are much smaller for now. But in two years' time, **AI revenues will start to hit single-digit %pts of GDP and become economically relevant**. At that point, we will need to see whether AI is mostly leading to efficiency gains (which is, at least initially, just a replacement for existing activity and jobs) or newer products/services and economic activity. The answer is likely both but if the mix is more towards the former to start with, economic activity may decrease elsewhere. To a certain extent, this started to be priced in the US software companies earlier this year, and in Indian IT stocks through the year. Eventually the freed up resources will need to be deployed into new products and services that we don't yet know of – and it is unclear how long this might take.

(2) **How will the mix share of value added from AI change**? The current setup is such that the hardware layer is earning the most profits. The hyperscaler/hosting layer is quite profitable but has vast up-front capex commitments. The model layer is the fastest-growing but with unclear profitability. This is very different from the internet and cloud era where software companies captured the vast majority of value, and hardware companies on average operated on thin margins. Value mix should shift in that direction. But AI businesses are not platform businesses of the internet era and do not benefit from the same network effects – and as such lack significant moats. Thus, the distribution of value-capture may end up being more balanced than (i.e. no layer earning disproportionate profits) and effectively operating on a cost+margin basis. This should leave the value creation from AI predominantly in the hands of the consumers. Similarly, even at the enterprise level, first movers in using AI may have a competitive advantage for a while (and be able to monetize that), but if everyone uses AI this advantage gets competed away and AI spend just becomes a cost item – leaving the resultant surplus with the AI companies and the end consumer.

(3) But the attractiveness of this consumer surplus is debatable if the end consumer is also being displaced from the labor market. There are some paths in which both of these phenomena taken to extremes lead to a world of abundance and human pursuits take other directions (such as leisure, spiritualism, higher-level enquiries, etc.). In other paths, this leads to social unrest and political upheaval. In either case, it is likely that **AI concentrates wealth and influence**, further exacerbating the K-shaped economy.

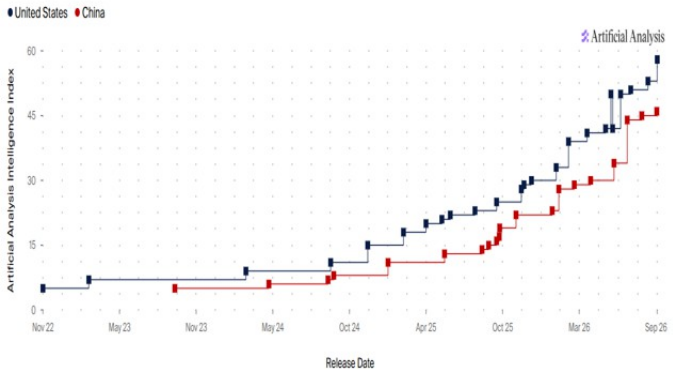
(4) Already starting to become salient this year, **regulation and AI-safety concerns** can affect deployment timing, revenue ramp, and model-release cadence. At this point, it does not look likely that frontier AI model training will slow (even if the rollout of new models is gated by alignment restrictions), but a serious incident can quickly change the nature of this debate. There are various vectors of risk here including: cybersecurity, data security, physical damage from embodied AI, emotional attachment to AIs, financial stability risks from agentic action, etc.

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(5) Perhaps the most straightforward of these issues is **geopolitics**. With the increasing capability of AI systems (as it pertains to cybersecurity, economic influence and potential military applications) and no end in sight, this is naturally becoming a geopolitical hot-button issue, with both the US and China focused on their respective AI ecosystems. Indeed, if AI is becoming anything as existential as defense or energy consumption, global semiconductor spending has a lot of room to grow.

Figure 48: A sizable portion of that gross profit will need to be spent on training/R&D to maintain lead – unless RSI is about to lower the R&D costs there

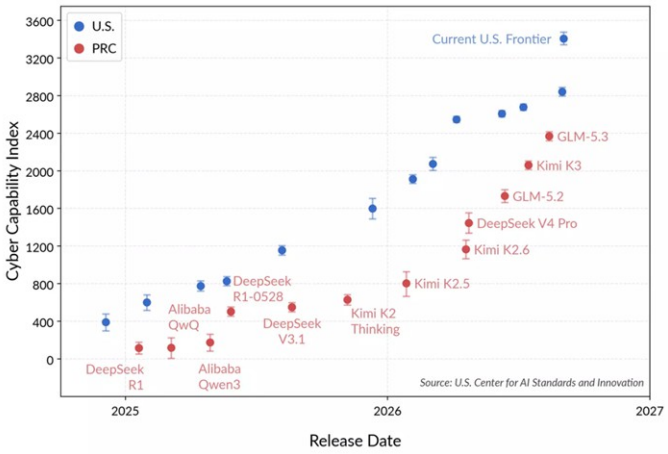
Intelligence index score



Source: Artificial Analysis.

Figure 49: Frontier capability progression (and US-China gap) is similar in the specific field of cyber capability

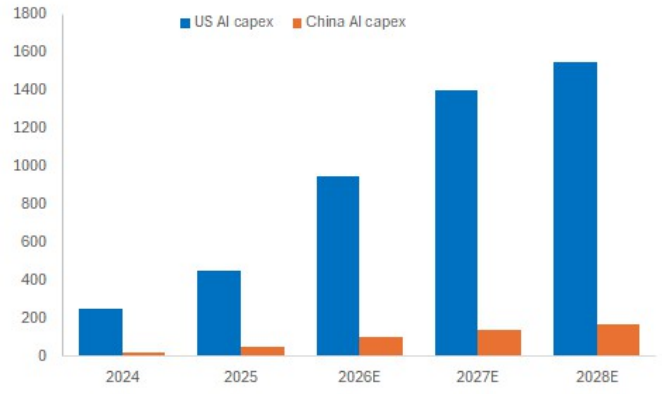
See here



Source: CAISI/NIST.

Figure 50: There is presently a large gap between the size of US vs China AI capex, but both are growing rapidly

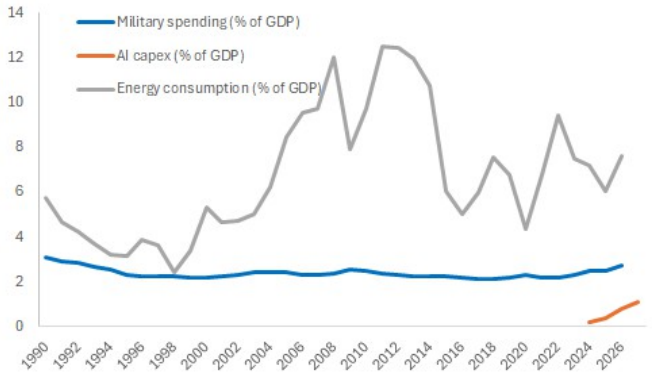
US estimates from JPMorgan Cloud Capex database. China estimates here



Source: Bloomberg Finance L.P., J.P. Morgan estimates.

Figure 51: If AI is as critical as defense and energy spending, there is a lot of room for global AI capex to rise

% of GDP



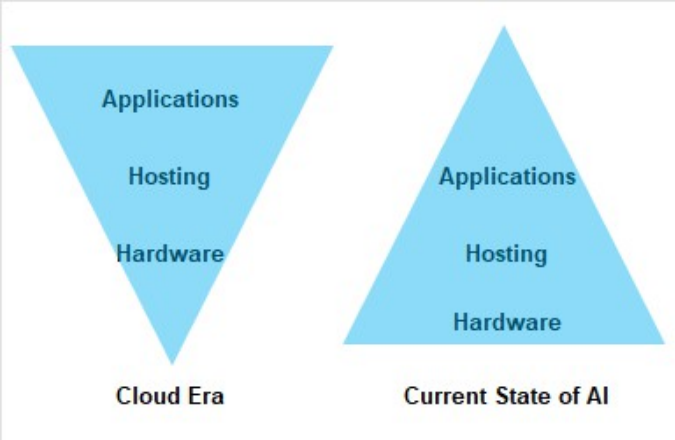
Source: Bloomberg Finance L.P., J.P. Morgan.

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Figure 52: Current value capture mix in AI looks very different from the cloud era

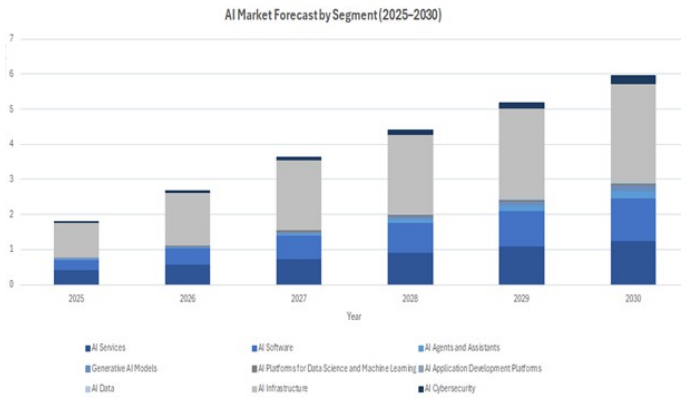
Indicative distribution of value capture across the stack



Source: J.P. Morgan.

Figure 53: AI driving incremental $430bn+ Security 3yr TAM

Market size estimate in US$ tn. See more here



Source: Gartner report.

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